

NoSQL MongoDB – Course Outline

Duration: 40 hours

Level: Intermediate

Course Description:

This course provides an in-depth understanding of NoSQL databases with a focus on MongoDB. Learners will explore data modeling, CRUD operations, indexing, aggregation, and MongoDB performance tuning. The course also covers practical applications of MongoDB in real-world scenarios.

Tools Used:

- MongoDB Server & MongoDB Atlas
- MongoDB Compass
- MongoDB Shell
- Python (for integration)

Prerequisites:

- Basic understanding of databases
- Familiarity with JSON structure
- Basic knowledge of Python or Node.js is recommended

Course Outcomes:

By the end of this course, learners will be able to:

- Understand NoSQL concepts and MongoDB architecture
- Design and model data for MongoDB databases
- Perform CRUD operations efficiently
- Utilize advanced queries, indexing, and aggregation frameworks
- Apply best practices for MongoDB deployment and performance tuning
- Integrate MongoDB with applications

Course Outline:

Module No.	Module Name	Module Descriptions	Hours
1	Introduction to NoSQL and MongoDB	• Difference between SQL and NoSQL • Types of NoSQL Databases • Advantages of MongoDB • MongoDB Architecture and Components • Installation and Setup (Local & Atlas)	6
2	MongoDB Basics and CRUD Operations	• Databases, Collections, and Documents • JSON and BSON Structure • Create, Read, Update, Delete (CRUD) • Data Types in MongoDB • Using MongoDB Compass for CRUD • MongoDB Shell and Command Line Operations	7
3	Data Modeling and Indexing	• Schema Design and Embedding vs Referencing • One-to-One, One-to-Many, Many-to-Many Relationships • Index Types: Single Field, Compound, Multikey • Indexing Best Practices • Performance Impact of Indexing	7
4	Advanced Queries and Aggregation	• Query Operators and Logical Operators • Projection, Sorting, Limiting • Aggregation Pipeline Stages • Grouping and Filtering Data • Using \$lookup for Joins • Real-Time Aggregation Examples	7
5	MongoDB Performance and Security	• Performance Tuning Techniques • Profiling Queries • Replication Concepts • MongoDB Sharding Basics • Authentication and Authorization • Role-Based Access Control	6
6	Integration and Case Studies	• Connecting MongoDB with Python (PyMongo) • CRUD Operations through Python • MongoDB Integration with Web Applications • Real-World Case Studies using MongoDB • Hands-on Project: Building a MongoDB-Backed Application	7

Conclusion:

This course equips learners with practical skills to work confidently with MongoDB in real-time applications. By mastering intermediate-level MongoDB concepts, learners will be capable of handling data-intensive projects and preparing for advanced NoSQL solutions in the industry.